

RNA-seq Data Analysis

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Roadmap of STAT/BIOF/GSAT540 2026

data generation followed by *inference* or validation

- ③ Formulating biological questions
- ④ Hypothesis testing
- ⑤ **RNA-seq data** (Today)
- ⑥ **Inference focusing on RNA-seq** (Thursday)
- ⑦ **Experimental design** (next week)
- ⑧ Single-cell genomics
- ⑨ Matrix factorization
- ⑩ Genome-wide association studies
- ⑪ Multiple hypothesis testing

Learning objectives of RNA-seq analysis

How do we get sequencing data?

- High-throughput sequencing data
 - data generation scheme
 - quantification
- Data normalization
 - intrinsic biases
 - gene-level adjustment
 - gene and sample adjustment
- Over-dispersed nature
 - count data
 - transformation

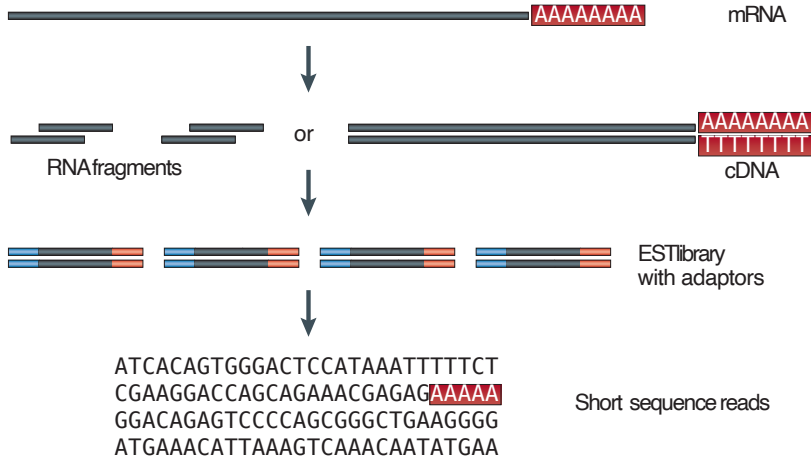
How can we fit a model to sequencing data?

- Linear modelling
 - a Gaussian linear model
 - a generalized linear model
- Multivariate linear models
 - a column space view
- Model-based inference
 - maximum likelihood learning
 - parametric inference
- Revisit RNA-seq
 - [DESeq2](#) vs. [limma-voom](#)
 - modelling and estimation

Today's lecture

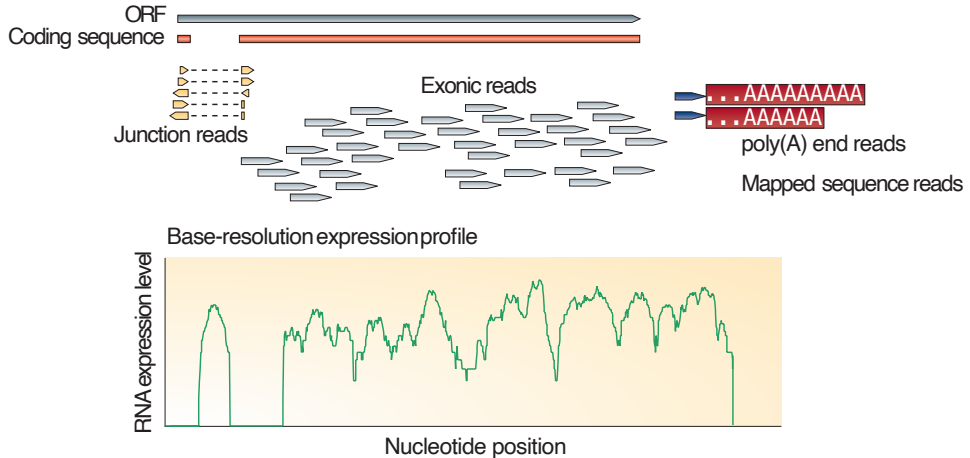
- 1 RNA-Seq data generation scheme
- 2 RNA-seq data normalization
- 3 Understanding the nature of RNA-seq data

RNA-seq: pooling mRNA reads with poly-A tails



Wang et al. (2009)

RNA-seq: mapping short reads to reference genome



Wang et al. (2009)

Before RNA-seq: The Pre-2008 Era

EST sequencing (1990s)

- Low throughput Sanger sequencing of cDNA clones
- Expensive and labor-intensive

SAGE (1995)

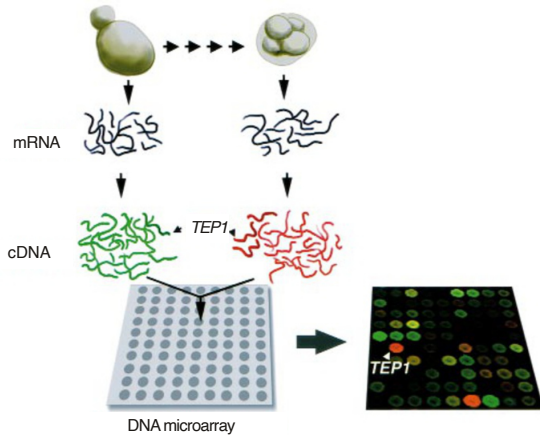
- Short sequence tags from cDNA
- Limited dynamic range

Microarrays (late 1990s-2000s)

- Required **predefined probes** for known transcripts
- High background noise, cross-hybridization
- Limited dynamic range (~3 orders of magnitude)
- Poor at detecting low-abundance transcripts
- Species-specific probe design required

Wang et al. (2009)

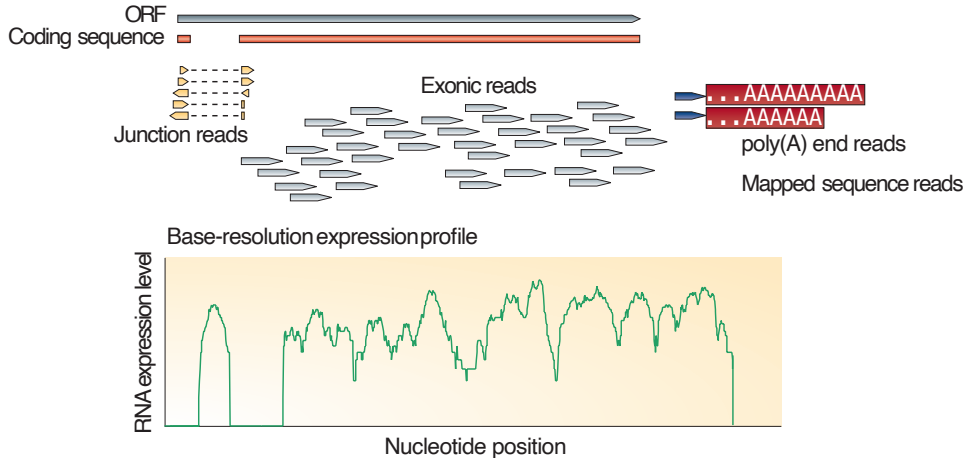
Microarrays had been revolutionary, too



*Thousands of genes are being discovered for the first time by sequencing the genomes of model organisms, **an exhilarating reminder** that much of the natural world remains to be explored at the molecular level.*

Brown and Botstein (1999)

RNA-seq: mapping short reads to reference genome



Wang et al. (2009)

Why sequencing-based high-throughput assays?

Key improvements

- Much wider dynamic range (~5 orders of magnitude)
- No prior annotation needed
- Single-nucleotide resolution
- Digital counts vs. analog signals
- Works for any organism

New capabilities

- Discovery of novel transcripts and genes
- Alternative splicing detection
- Allele-specific expression
- Study of non-model organisms
- Precise isoform quantification

From **hypothesis-driven** (design probes; fixed arrays)



discovery-driven (sequence everything, analyze later)

Foundational experimental work (2008)

Nagalakshmi et al. (2008)

- First complete transcriptome map using deep sequencing
- Yeast genome transcriptional landscape

Mortazavi et al. (2008)

- First comprehensive RNA-seq in mouse tissues
- Introduced **RPKM** normalization
- Demonstrated superiority over microarrays

Sequencing Technologies

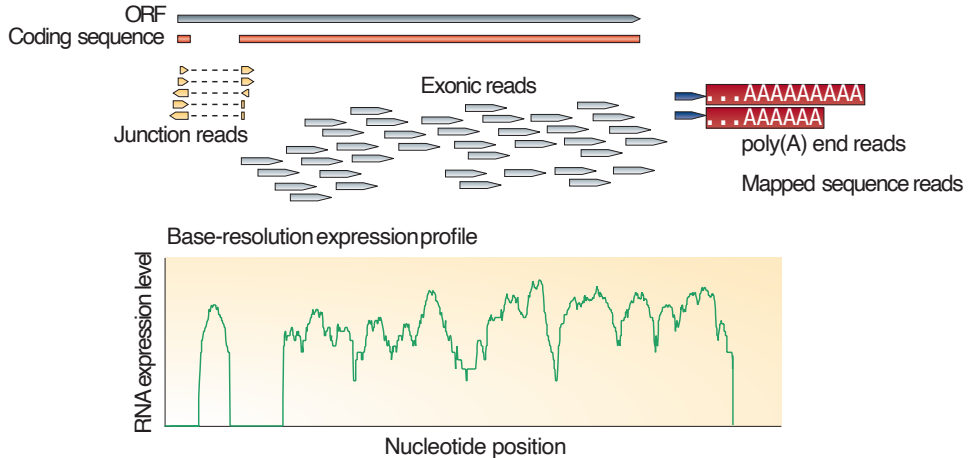
Early Platforms (2005-2010)

- **454 Pyrosequencing:** First NGS platform, longer reads but lower throughput
- **Illumina/Solexa:** Became dominant platform, higher throughput, shorter reads
- **SOLiD:** Two-base encoding sequencing

Modern Era (2011-present)

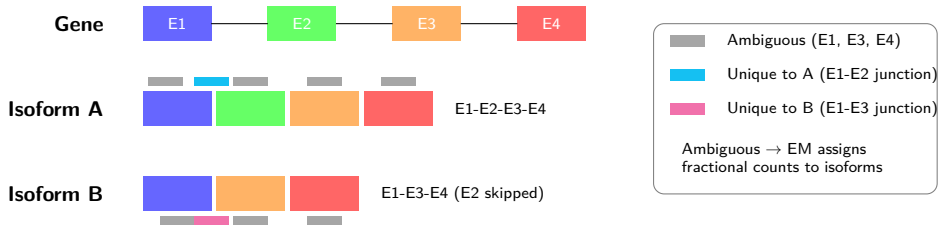
- **Pacific Biosciences (PacBio):** Long-read sequencing, full-length transcripts without assembly
- **Oxford Nanopore:** Portable, long-read, direct RNA sequencing without reverse transcription

RNA-seq: mapping short reads to reference genome



Wang et al. (2009)

The isoform quantification problem



Most reads are **ambiguous** → require **statistical inference** (EM) to estimate isoform abundance

Long-read RNA-seq: Full-length transcripts

The isoform problem

- Short reads (~150bp) cannot span full transcripts
- Transcript assembly is computationally challenging
- Isoform quantification relies on statistical inference



Long-read solution: Sequence entire transcripts (1-10+ kb) without fragmentation

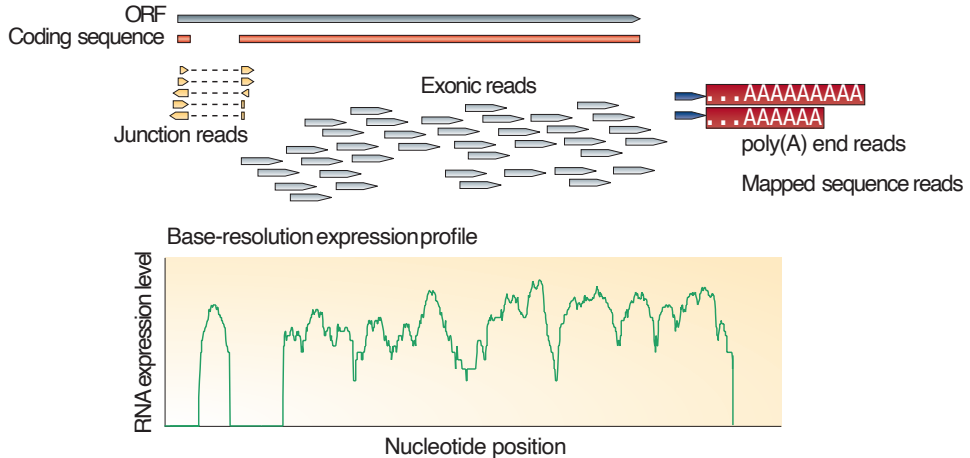
Key advantages

- Direct isoform identification without assembly
- Novel splice variant discovery
- Fusion transcript detection
- Allele-specific (break the tie)

Trade-offs

- Lower throughput than short-read
- Higher error rates
- Higher cost per base

RNA-seq: mapping short reads to reference genome



Wang et al. (2009)

Gene-level quantification methods (CS-oriented)

Burrow-Wheeler (2009-2012)

- **bowtie** (Langmead et al. 2009): Ultrafast “dictionary” based mapping
- **TopHat** (Trapnell et al. 2009): Parsing out the bowtie results
- **cufflinks** (Trapnell et al. 2010): Exact transcript assembly model (first suggested FPKM)
- **RSEM** (Li and Dewey 2011): Expectation Maximization-based splice isoform quantification

Isoform-aware mapping (2012-2014)

- **STAR** (Dobin et al. 2013): longer reads and splice-aware

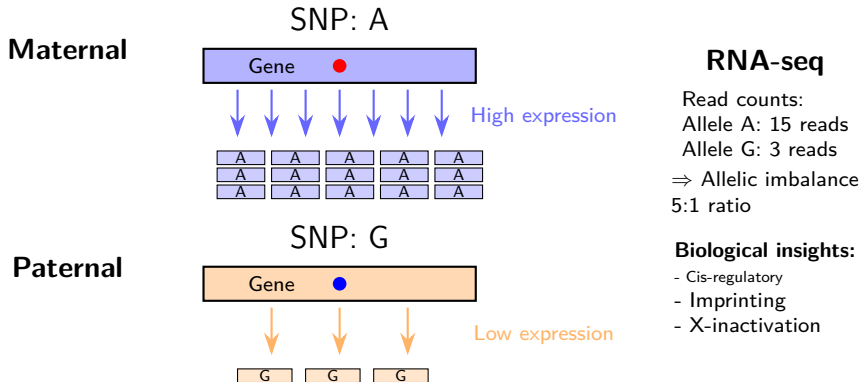
Pseudo-alignment (2015-2017)

- **kallisto**: Ultra-fast pseudoalignment, orders of magnitude faster (Bray et al. 2016)
- **salmon**: Fast, bias-aware quantification (Patro et al. 2017)

Hands-on session in Seminar 06

RNA-seq: from beginning to end – BAM to count data

Aside: Allele-specific analysis separately quantifies diploid transcriptome



(Montgomery et al. 2010; Pickrell et al. 2010) Let's do this in 1ect12

Today's lecture

- 1 RNA-Seq data generation scheme
- 2 RNA-seq data normalization**
- 3 Understanding the nature of RNA-seq data

Why normalize RNA-seq data?

Technical biases

- Sequencing depth varies across samples
- Gene length affects read counts
- GC-content bias
- Batch effects

Goal

- Make expression levels comparable across genes and samples
- Remove technical variation while preserving biological signal

Normalization methods

Within-sample normalization

- **RPKM/FPKM**: Reads per kilobase per million mapped reads (Mortazavi et al. 2008)
- **TPM**: Transcripts per million (improved RPKM)

Adjusts for sequencing depth and gene length

Between-sample normalization

- **TMM**: Trimmed mean of M-values (Robinson and Oshlack 2010)
- **DESeq size factors**: Median of ratios (Anders and Huber 2010)
- **RUVSeq**: Removing unwanted variation (Risso et al. 2014)

Addresses composition bias

Continue to work on TCGA data

```
data.dir <- "../data/recount3"
dir.create(data.dir, recursive = TRUE, showWarnings = FALSE)
.file <- file.path(data.dir, "tcga_brca_rse.rds")

if.needed(.file, {
  tcga.proj <- subset(available_projects(), file_source ==
    "tcga")
  brca.info <- tcga.proj[tcga.proj$project == "BRCA", ]
  rse.brca <- create_rse(brca.info)
  saveRDS(rse.brca, .file)
})
rse <- readRDS(.file)
```

After alignment: SummarizedExperiment

		samples (columns)						
		+-----+						
			S1	S2	S3	...		
		+-----+	+-----+					
genes (rows)	G1		10	25	18	...		<- assays (counts)
	G2		5	12	8	...		
	...		...	...	...	...		
		+-----+	+-----+					

rowData: genomic coordinates (chr, start, end, strand)

colData: sample metadata (tumor/normal, clinical info)

Learning objectives: identify biases and address them

- ① Gene bias (rows)
- ② Sequencing depth (samples/columns)
- ③ Composition bias (corollary to sequencing depth)

Get the data and compute the statistics

Here is gene expression count data:

```
raw.counts <- assay(rse, "raw_counts")
```

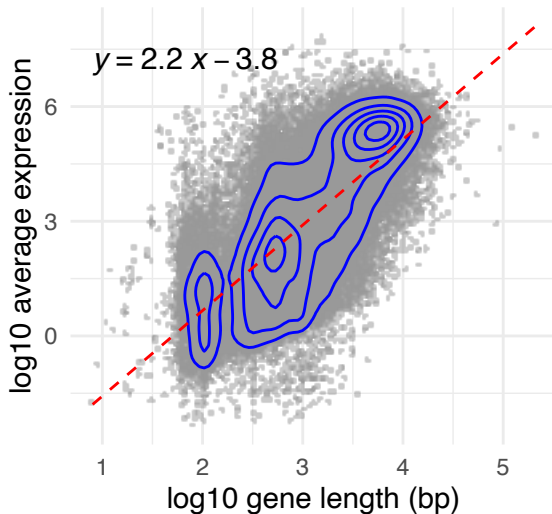
Here is gene/row annotations

```
rowData(rse) %>%  
  head(3)
```

```
## DataFrame with 3 rows and 10 columns
```

```
##           source      type bp_length      phase      gene_id  
##           <factor> <factor> <numeric> <integer>    <character>  
## ENSG00000278704.1 ENSEMBL   gene      2237        NA ENSG00000278704.1  
## ENSG00000277400.1 ENSEMBL   gene      2179        NA ENSG00000277400.1  
## ENSG00000274847.1 ENSEMBL   gene      1599        NA ENSG00000274847.1  
##           gene_type  gene_name      level havana_gene  
##           <character> <character> <character> <character>  
## ENSG00000278704.1 protein_coding BX004987.1      3      NA  
## ENSG00000277400.1 protein_coding AC145212.2      3      NA  
## ENSG00000274847.1 protein_coding AC145212.1      3      NA  
##           tag  
##           <character>
```

Bias 1. Gene length vs. gene expression



How did the gene length bias occur?

Original transcript:

5' -- [5' UTR] -- [=====CDS=====] -- [3' UTR] -- AAAAAAA 3'
(polyA)

RNA is fragmented before sequencing

After fragmentation:

==== ==== ==== ==== ==== ==== ==
 ==== ==== ==== ==== ==

Longer genes produce more fragments

Short gene (1 kb):

|-----|
== == ==
== ==
~5 reads

Long gene (4 kb):

|-----|
== == == == == == == == == ==
== == == == == == == == ==
~20 reads

Same expression $\Rightarrow$ $4 \times$ more counts for the $4 \times$ longer

RPKM (Reads Per Kilobase per Million mapped)

$$\text{RPKM}_{gi} = \frac{Y_{gi}}{L_g/1000 \times N_i/10^6} = \frac{Y_{gi} \times 10^9}{L_g \times N_i}$$

Where:

- Y_{gi} = raw count for gene g in sample i
- L_g = gene length of gene g in bp
- N_i = total mapped reads on a sample i (library size)

Order: Normalize by library size first, then gene length.

Mortazavi et al. (2008)

TPM (Transcripts Per Million)

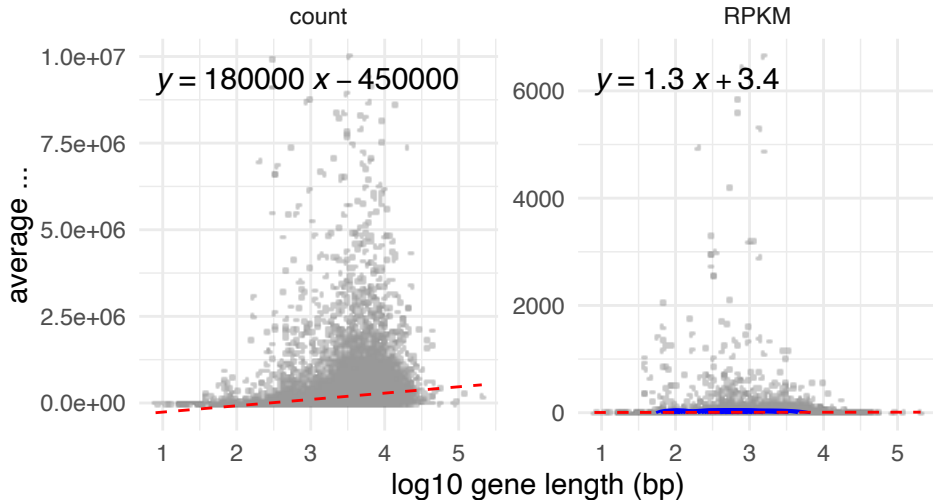
$$\text{TPM}_{gi} = \frac{Y_{gi}/L_g}{\sum_{g'} (Y_{g'i}/L_{g'})} 10^6 = \frac{\text{FPKM}_{gi}}{\sum_{g'} \text{FPKM}_{g'i}/10^6}$$

Order: Normalize by gene length first, then scale to per million.

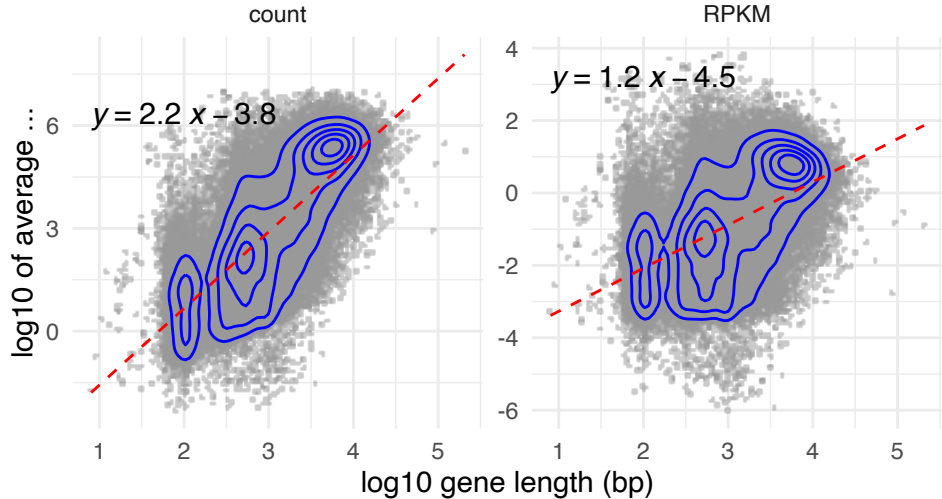
Key difference: TPM values sum to 1 million across genes within each sample, making samples directly comparable.

Li and Dewey (2011), Wagner et al. (2012)

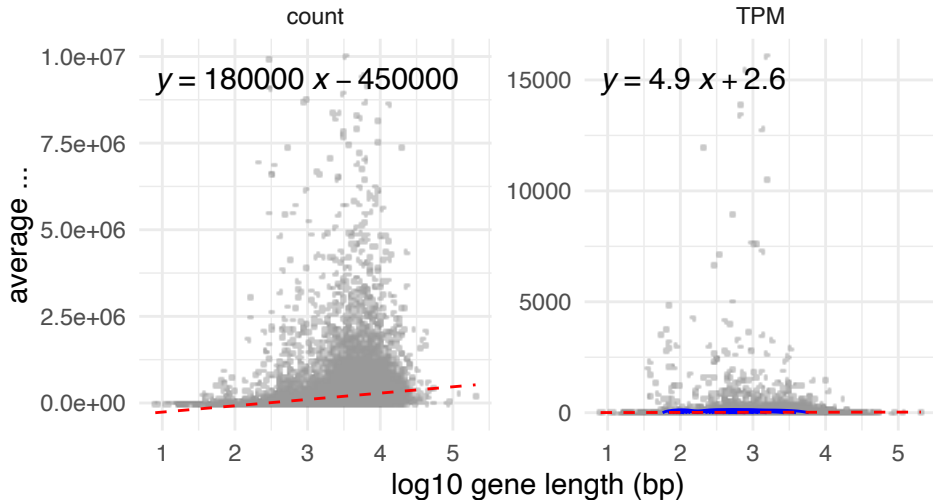
Gene length vs. expression after RPKM adjustment



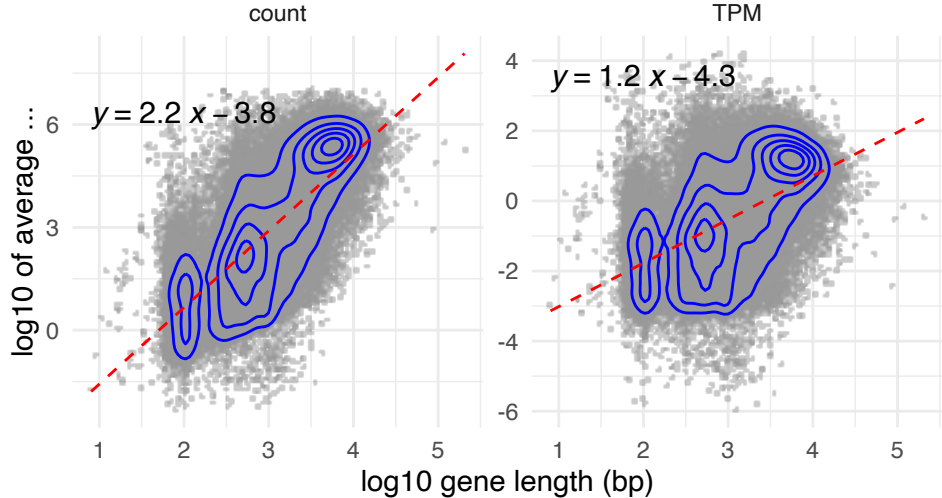
Gene length vs. expression after RPKM adjustment



Gene length vs. expression after TPM adjustment



Gene length vs. expression after TPM adjustment

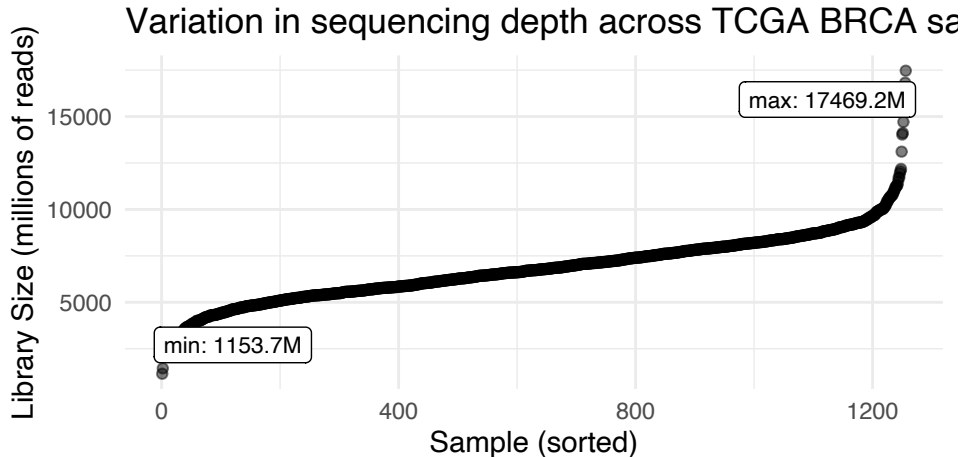


Bias 2. Sequencing depth and library size

- Different samples are sequenced to different depths
- Different total read counts even when biological content is identical.

```
lib.sizes <- colSums(raw.counts)
```

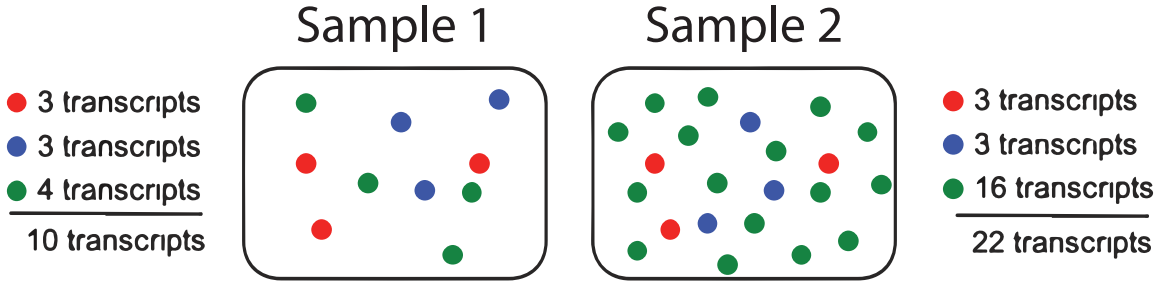
Bias 2. Sequencing depth and library size



Bias 3. RNA composition change by normalization

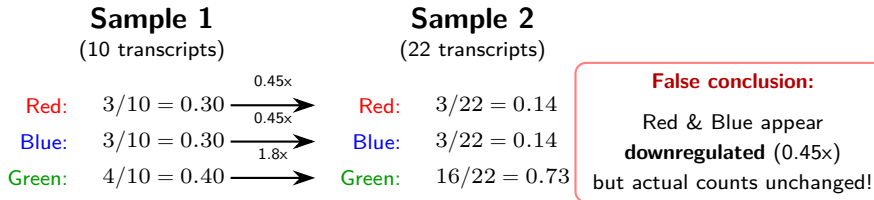
- Two samples, each with 1 million total reads
- Sample A: 1000 genes each expressed equally (1000 reads/gene)
- Sample B: One gene upregulated $100\times$, now takes 500,000 reads
 - The remaining 999 genes now share only 500,000 reads (~ 500 reads/gene)
 - These 999 genes appear “downregulated” even though their absolute expression is unchanged!

RNA composition bias due to outlier genes



What would happen if normalized by library size?

What would happen with naive normalization



Problem: Green gene's high expression "steals" proportional representation from others

TMM (Trimmed Mean of M-values) by Robinson and Oshlack (2010)

- “A scaling normalization method for differential expression analysis of RNA-seq data”
- Implemented in **edgeR** Bioconductor package
- Calculate scaling factors based on weighted trimmed mean of log-fold-changes between samples.
- Normalization tailored for differential expression analysis (Thursday)

TMM algorithm

- 1 Choose a reference sample (typically sample with median library size)
- 2 For each sample vs. reference, calculate:
 - M-values (log fold-change):

$$M_g = \log_2 \left(\frac{\text{count}_g^{\text{sample}}}{\text{count}_g^{\text{ref}}} \right)$$

- 3 Trim extreme M-values (default: 30%)
- 4 Calculate weighted mean of remaining M-values
- 5 Convert to scaling factor

Assumption: Most genes are not differentially expressed between samples.

TMM implemented in edgeR

```
library(edgeR)
```

```
## Loading required package: limma
```

```
##
```

```
## Attaching package: 'limma'
```

```
## The following object is masked from 'package:BiocGenerics':
```

```
##
```

```
##      plotMA
```

```
## Create DGEList object
```

```
dge <- DGEList(counts = raw.counts)
```

```
## Calculate TMM normalization factors
```

```
dge <- calcNormFactors(dge, method = "TMM")
```

Another idea implemented in DESeq2

```
library(DESeq2)
coldata <- data.frame(sample = colnames(raw.counts))

dds <- DESeqDataSetFromMatrix(countData = raw.counts, colData = coldata,
                              design = ~1)
```

```
## converting counts to integer mode
```

```
## Calculate size factors
```

```
dds <- estimateSizeFactors(dds)
```

```
## Get normalized counts
```

```
deseq.norm <- counts(dds, normalized = TRUE)
```

Summary: Three main sources of bias

1. **Gene length bias** (RPKM/FPKM/TPM)
2. **Sequencing depth / Library size bias**
3. **RNA composition bias** (TMM)

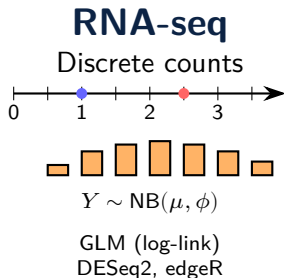
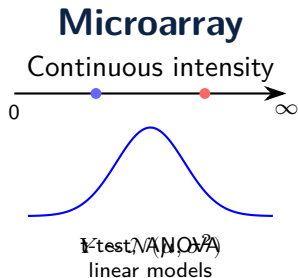
Pop quiz!

- ❶ For differential expression analysis, namely gene-by-gene comparison, which normalization methods should be applied?
- ❷ For gene set enrichment analysis, which normalization methods should be applied?

Today's lecture

- 1 RNA-Seq data generation scheme
- 2 RNA-seq data normalization
- 3 Understanding the nature of RNA-seq data**

Sequencing generates count data



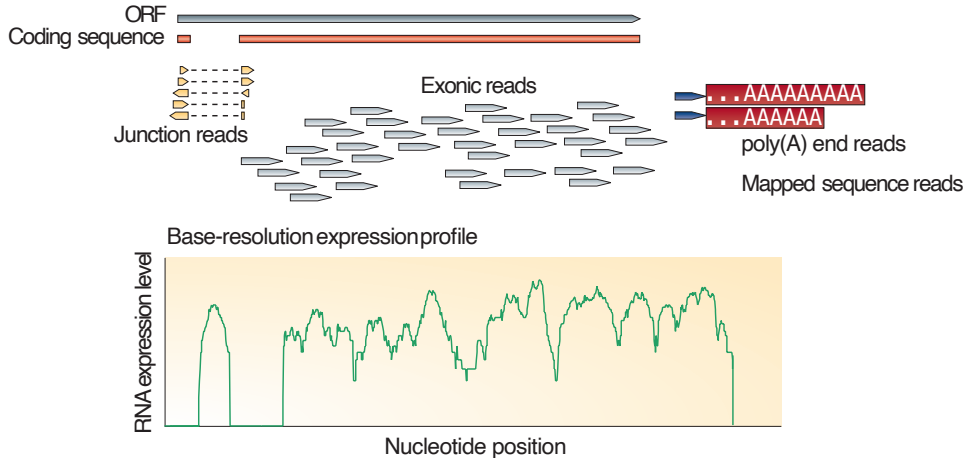
Microarray

- Continuous intensity values
- Normal distribution assumptions

RNA-seq

- Discrete count data
- Poisson/Negative binomial models

RNA-seq: mapping short reads to reference genome



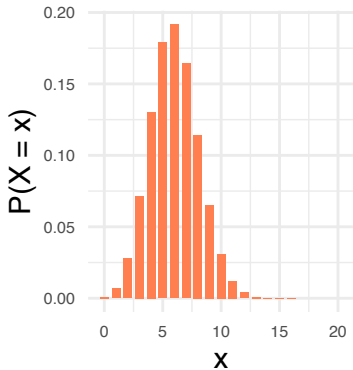
Wang et al. (2009)

Learning objectives

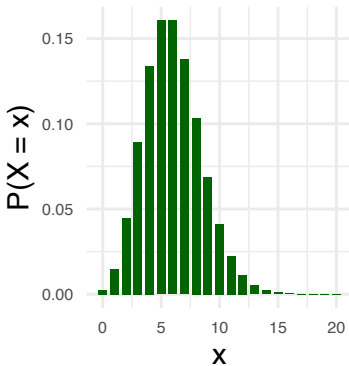
- ① Understand mean-variance relationship in RNA-seq
- ② Why Negative Binomial distribution for RNA-seq
- ③ Identify challenges in NB modelling

Recap: the distributions for discrete data

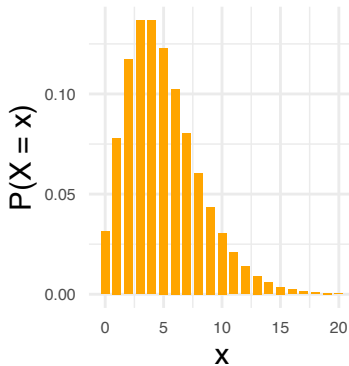
Binomial($n=20, p=0.3$)



Poisson($\lambda=6$)



Negative Binomial
($r=5, p=0.5$)



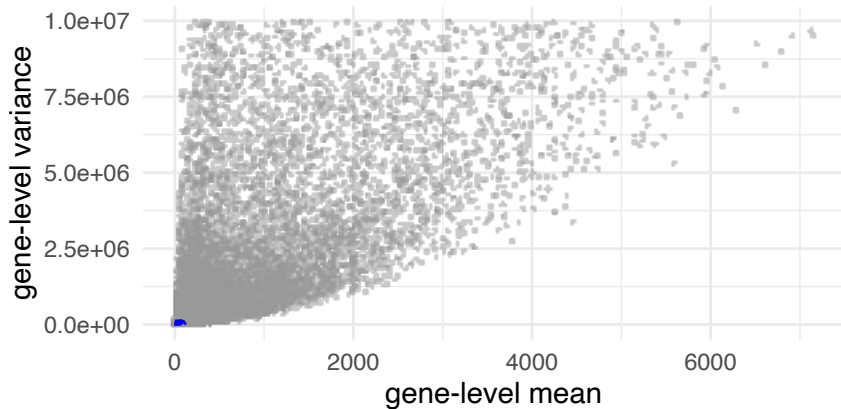
Which one best captures RNA-seq count data?

Let's take a look at gene-level statistics

```
## Calculate mean and variance per gene  
gene.mean <- rowMeans(raw.counts)  
gene.var <- rowVars(raw.counts)
```

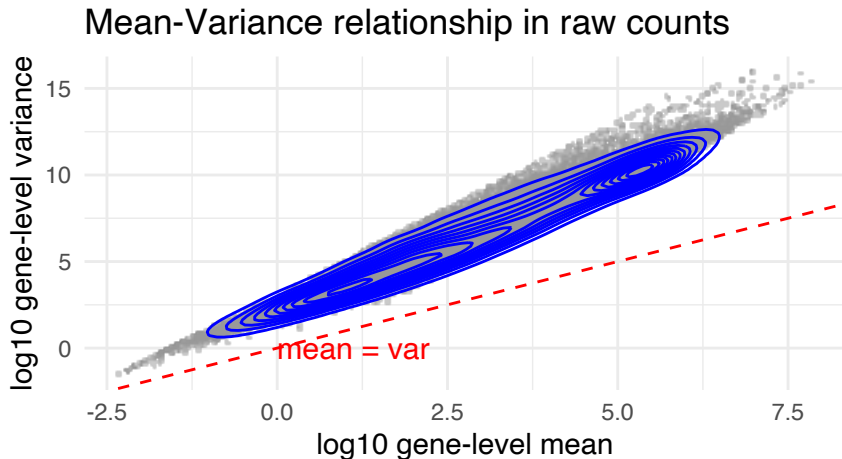
Mean-variance relationships in RNA-seq data

Mean-Variance relationship in raw counts



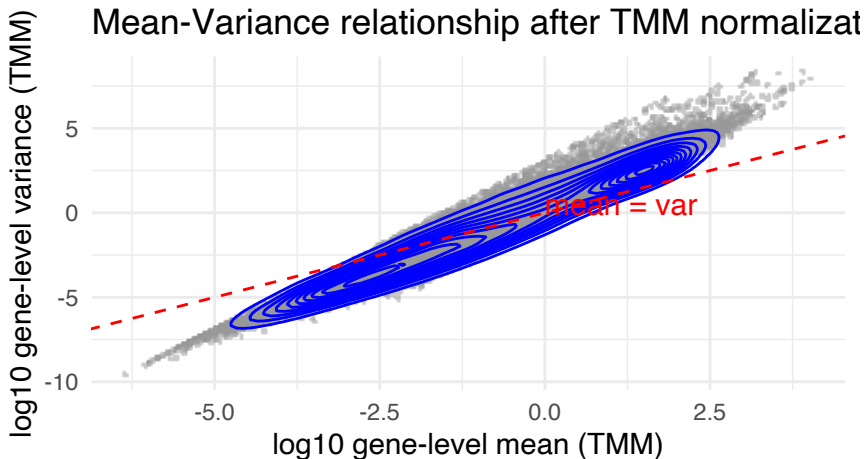
What is this called?

Mean-variance relationships in RNA-seq data



What is this called? **overdispersion**

Does TMM normalization fix overdispersion?



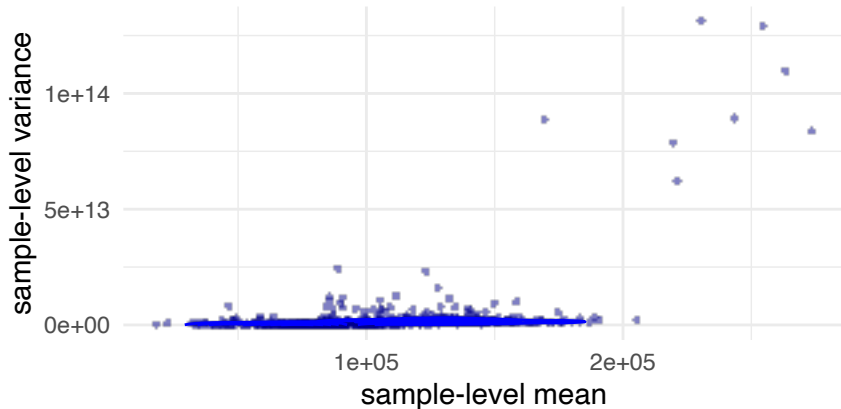
TMM addresses **composition bias**, not **overdispersion**.

What about column-wise mean-variance?

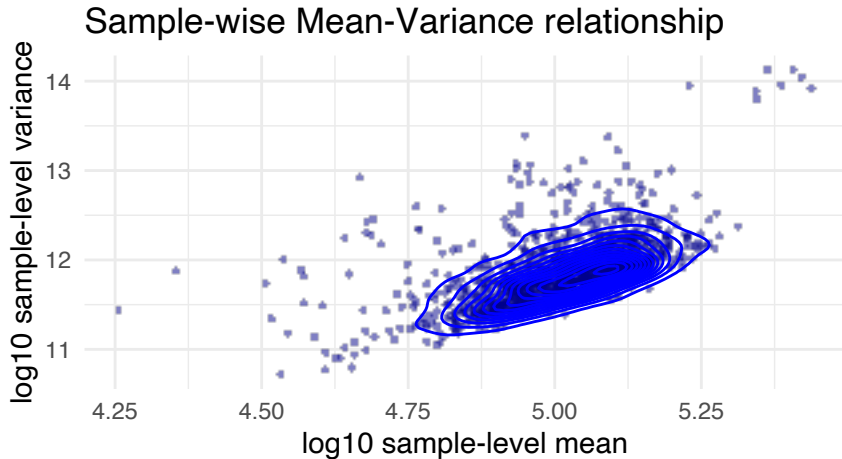
```
sample.mean <- colMeans(raw.counts)
sample.var <- colVars(raw.counts)
```


Mean and variance per sample (column-wise)

Sample-wise Mean-Variance relationship

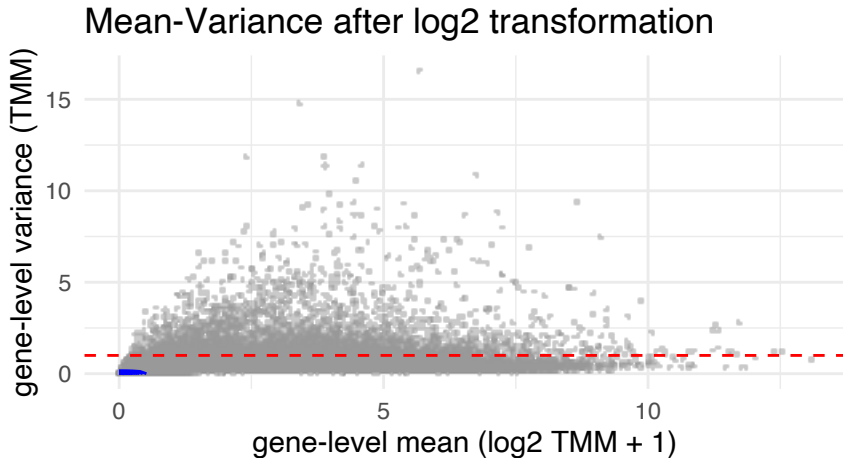


Mean and variance per sample (column-wise)



similar mean-variance relationship?

What about log2-transformed gene-level variance?



Log transformation helps, but **low-expression genes still have high variance**

What does “over” dispersion of a random variable mean?

- Is there “under” dispersion?
- If there is neither “over” nor “under” dispersion:

$$\mathbb{E}[X] = \mathbb{V}[X]$$

Which one for RNA-seq data?

$$\mathbb{E}[X] < \mathbb{V}[X]$$

$$\mathbb{E}[X] > \mathbb{V}[X]$$

Why not Poisson?

Poisson assumption

$$\text{Var}(Y) = \mu$$

- Mean equals variance
- Too restrictive for biological data

Biological reality

$$\text{Var}(Y) > \mu$$

- Overdispersion is common
- Variability exceeds expectation

Negative Binomial model

$$Y \sim \text{NB}(\mu, \phi)$$

- μ : mean expression
- ϕ : dispersion parameter (captures overdispersion)
- $\phi > 0$

$$\mathbb{V}[Y] = \mu + \phi\mu^2$$

Negative Binomial model

- μ : mean expression
- ϕ : dispersion parameter (captures overdispersion)
- $\phi > 0$

$$P(Y = y) = \binom{y + r - 1}{y} p^r (1 - p)^y$$

where $r = 1/\phi$ and $p = 1/(1 + \phi\mu)$

$$\mathbb{V}[Y] = \mu + \phi\mu^2$$

Negative Binomial model

- μ : mean expression
- ϕ : dispersion parameter (captures overdispersion)
- $\phi > 0$

$$\mathbb{V}[Y] = \mu + \phi\mu^2$$

$$P(Y = y) = \frac{\Gamma(y + 1/\phi)}{\Gamma(1/\phi) \Gamma(y + 1)} \left(\frac{1}{1 + \phi\mu} \right)^{1/\phi} \left(\frac{\phi\mu}{1 + \phi\mu} \right)^y$$

where $\Gamma(\cdot)$ is Euler's gamma function.

Some intuition of NB distribution

Classical interpretation

NB models the number of **failures** before achieving r **successes** in a sequence of independent Bernoulli trials.

- r : target number of successes
- p : probability of success in each trial
- Y : number of failures before r successes

Special cases

$r = 1 \Rightarrow$ **Geometric distribution**

- Number of failures before the first success

$r \rightarrow \infty$ (or $\phi \rightarrow 0$) $\Rightarrow$ **Poisson**

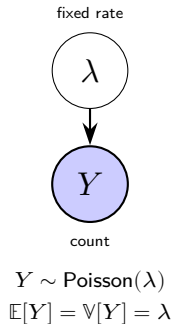
- Overdispersion vanishes
- Mean = Variance

For RNA-seq:

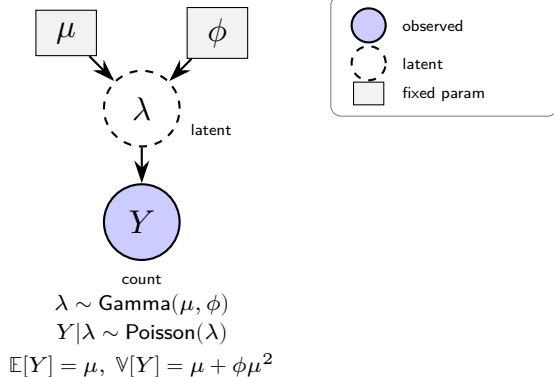
- NB is a flexible count distribution
- Accommodates overdispersion through ϕ (or r)

Poisson vs. Negative Binomial in terms of μ, ϕ

Poisson

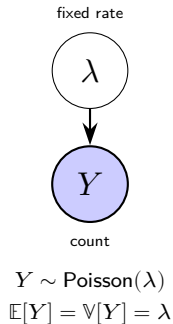


Negative Binomial

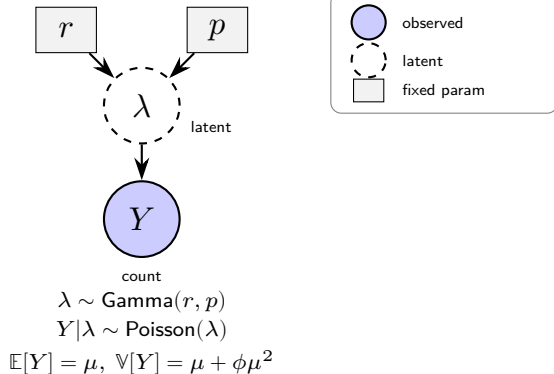


Poisson vs. Negative Binomial

Poisson



Negative Binomial



Optional: different parameterizations of NB

(r, p) parameterization

$$Y \sim \text{NB}(r, p)$$

$$\mathbb{E}[Y] = \frac{r(1-p)}{p}$$

$$\mathbb{V}[Y] = \frac{r(1-p)}{p^2}$$

(μ, ϕ) parameterization

$$Y \sim \text{NB}(\mu, \phi)$$

$$\mathbb{E}[Y] = \mu$$

$$\mathbb{V}[Y] = \mu + \phi\mu^2$$

Optional: different parameterizations of NB

$$\mu = \frac{r(1-p)}{p} \quad \phi = \frac{1}{r}$$

$$r = \frac{1}{\phi} \quad p = \frac{1}{1 + \phi\mu}$$

As $\phi \rightarrow 0$ (or $r \rightarrow \infty$): NB $\rightarrow$ Poisson with mean μ

Aside: Why Gamma-Poisson = Negative Binomial?

Hierarchical model:

$$\lambda|r, \beta \sim \text{Gamma}(r, \beta), \quad Y|\lambda \sim \text{Poisson}(\lambda)$$

Marginal distribution (integrate out λ):

$$\begin{aligned} P(Y = y|r, \beta) &= \int_0^\infty P(Y = y|\lambda) P(\lambda) d\lambda \\ &= \int_0^\infty \frac{\lambda^y e^{-\lambda}}{y!} \cdot \frac{\beta^r}{\Gamma(r)} \lambda^{r-1} e^{-\beta\lambda} d\lambda \\ &= \frac{\beta^r}{y! \Gamma(r)} \int_0^\infty \lambda^{y+r-1} e^{-(1+\beta)\lambda} d\lambda \end{aligned}$$

Aside: Why Gamma-Poisson = Negative Binomial?

Hierarchical model:

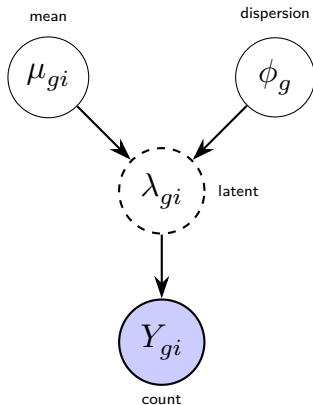
$$\lambda|r, \beta \sim \text{Gamma}(r, \beta), \quad Y|\lambda \sim \text{Poisson}(\lambda)$$

Marginal distribution (integrate out λ):

$$\begin{aligned} P(Y = y|r, \beta) &= \int_0^\infty P(Y = y|\lambda) P(\lambda) d\lambda \\ &= \frac{\beta^r}{y! \Gamma(r)} \cdot \frac{\Gamma(y+r)}{(1+\beta)^{y+r}} = \binom{y+r-1}{y} \left(\frac{\beta}{1+\beta}\right)^r \left(\frac{1}{1+\beta}\right)^y \end{aligned}$$

$\Rightarrow$ negative binomial PMF with $p = \beta/(1+\beta)$

Challenge of using NB distribution



$$\lambda_{gi} \sim \text{Gamma}(\mu_{gi}, \phi_g)$$
$$Y_{gi} | \lambda_{gi} \sim \text{Poisson}(\lambda_{gi})$$

- μ_{gi} : gene g , sample i
- ϕ_g : gene g only
- Y_{gi} : observed count

How many ϕ_g values?

- Just one: $\phi_g = \phi_{g'}$ for all $g \neq g'$
- For each gene: $\phi_g \neq \phi_{g'}$ if $g \neq g'$

Discussion

- ❶ Can I ignore over-dispersion if I don't have deep sequencing?
- ❷ I forgot to perform TMM normalization, but can I adjust library depth by incorporating overdispersion parameters?

Statistical methods for estimating overdispersion ϕ_g

- **edgeR**: Robinson and Smyth (2008)
- **DESeq**: Anders and Huber (2010)
- **DESeq2**: Love et al. (2014)

Next lecture

- ① How to estimate overdispersion ϕ_g
- ② How to incorporate our working model into μ_{gi}
- ③ Practical solutions using **DESeq2** and **limma-voom**

Summary

- RNA-seq count data generation process
- Potential Biases in RNA-seq data (row & column)
- Properties in RNA-seq-derived count data

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